
Deep Hedging under Black–Scholes and Heston Dynamics

Recover the known hedge first; then diagnose what changes under stochastic volatility

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Can terminal-error training recover the known finite-grid hedge—and remain useful when volatility and the traded instrument set become more complex?

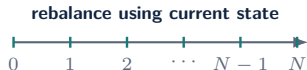
The hedge is learned through terminal portfolio error

$$\text{HE} = \pi + \underbrace{\sum_{n=0}^{N-1} \delta_n (S_{t_{n+1}} - S_{t_n})}_{\text{self-financing gains}} - (S_T - K)^+,$$
$$\min_{\pi, \theta} \mathbb{E}[\text{HE}^2].$$

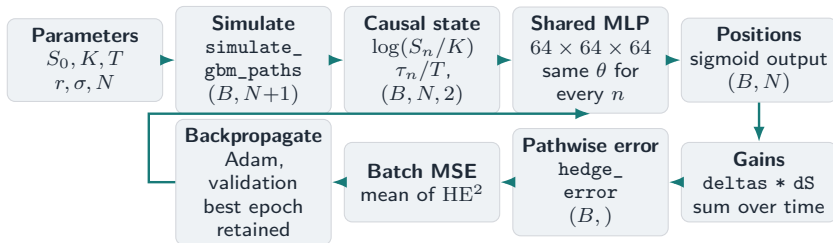
The analytic delta is never used as a training target.

Source: Submitted report, Secs. 1.1, 2.2.2 and 3.1; Eqns. (1.1)–(1.2).

- ▶ $N = 125$ discrete hedge dates in the baseline.
- ▶ $\delta_n = f_\theta(x_n)$ is adapted to information at t_n .
- ▶ Negative HE is a seller shortfall; reported seller loss is $L = -\text{HE}$.
- ▶ With $r = 0$, the submitted experiments use undiscounted gains.



The implementation is differentiable end to end

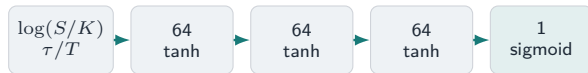


Supervision is financial: payoff minus self-financing wealth.

Boundary: Function names and path construction are verified from report Appendix C; the separate executable notebook bundle is absent from this repository.

Source: Submitted report, Algorithm A.6.1 and Listings C.1–C.3. Shape contract follows the displayed array construction and shared network input.

Call economics guide architecture selection



one shared map at all 125 dates

- ▶ **Log-moneyness** is scale-normalised and financially natural.
- ▶ **Weight sharing** exploits Markov and time-homogeneous structure.
- ▶ **Sigmoid** enforces the call-delta range $0 < \delta < 1$.
- ▶ A scalar premium is learned jointly with the hedge.

Candidate	Test RMSE	Time (s)
Shared norm. 64×3 , tanh	0.007891	103.0
Shared norm. 64×2 , ReLU	0.007912	93.2
Time-separated 16×1	0.008083	786.9

Selected on mean validation loss across three seeds; close alternatives remained.

Boundary: The model was selected, not uniquely dominant: the ReLU candidate won one seed.

Source: Submitted report, Secs. 3.1.2–3.1.3; Tables 3.1 and A.3–A.6.

The fair benchmark matches the grid and objective

No hedge

raw option risk

Black–Scholes delta

continuous-time rule sampled on the grid

Discrete-time MSE-optimal

conditional projection on the same grid

Degree-2 polynomial

deliberately low-capacity approximation

Neural hedge

terminal-MSE training without delta labels

same held-out paths

$$\delta_n^{\text{DT}} = \frac{\text{Cov}(g(S_T), \Delta S_n \mid S_{t_n})}{\text{Var}(\Delta S_n \mid S_{t_n})}$$

The comparison asks whether the network recovers the finite-grid solution—not whether it beats correct Black–Scholes theory.

Source: Submitted report, Sec. 3.1.1 and Appendix A.4; Schweizer (1995, 2001).

The neural hedge recovers the finite-grid benchmark

Strategy	RMSE
No hedge	0.227 257
Black–Scholes delta	0.007 810
Discrete-time MSE-optimal	0.007 810
Degree-2 polynomial	0.022 183
Neural hedge	0.007 841

Boundary: These are final-benchmark values from an independent retraining, not architecture-search rows.

0.4%

neural RMSE above the discrete-time
benchmark

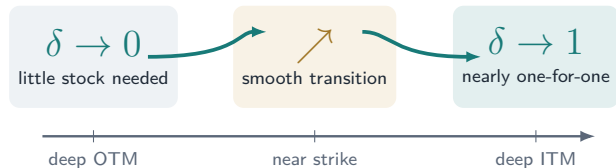
0.163972

learned premium
vs. analytic 0.164111

**Near equality is the
validation result we wanted.**

Source: Submitted report, Table 3.2 and Sec. 3.1.4.

The learned hedge has the right financial shape



- ▶ monotone in moneyness;
- ▶ close to both analytic curves;
- ▶ nearly indistinguishable on representative OTM, ATM and ITM paths.

The network learned the expected hedge function, not only a low aggregate loss.

Source: Economic reading of submitted report Figures 3.3 and 3.5; schematic above is not a data redraw.

Parameter Robustness under Retraining

Same architecture, **retrained from scratch** at each setting (one factor at a time) around baseline
 $(K, \sigma, T, N) = (0.9, 0.4, 0.5, 125)$.

$K \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$ $\sigma \in \{0.2, 0.4, 0.6\}$ $T \in \{0.25, 0.5, 1.0\}$ $N \in \{30, 60, 125, 250\}$

1.049

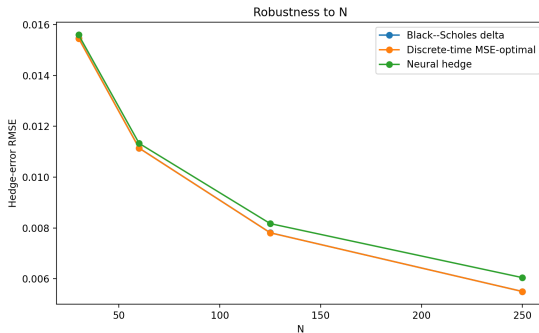
Mean NN/DT ratio (range 1.009–1.099)

$\leq 10\%$

Above DT-optimal, across all tested settings

$N = 250$

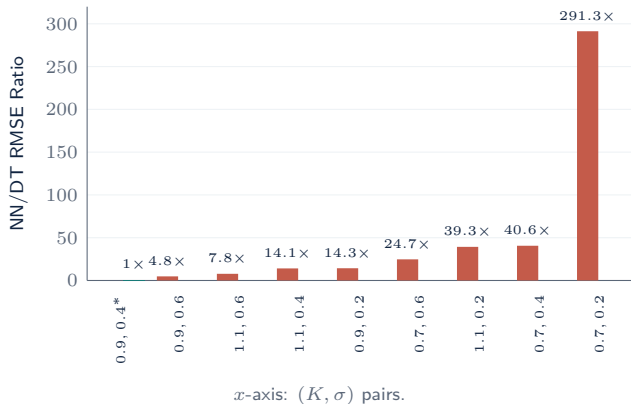
Worst case: benchmark shrinks faster than the network
here



Source: Submitted report, Sec. 4.1.1, Table 4.1 and Figure 4.1.

Generalization Failure of the Single-Scenario Hedger

One model **trained once** at $(K^*, \sigma^*) = (0.9, 0.4)$, evaluated elsewhere **without retraining**.



Why it fails

- 1. One scenario.** Trained on a single fixed (K^*, σ^*) — no variation to learn parameter dependence from.
- 2. Strike hardcoded.** K^* is baked into log-moneyness, so a new strike feeds the network the *wrong* number, not merely an unseen one.
- 3. No σ input.** Volatility is absent, so the network can't detect that market conditions have changed — it just reapplies whatever it learned at σ^* , regardless of the true volatility.

Source: Submitted report, Sec. 4.1.2, Table 4.2.

Conditioning fixes the Generalisation Failure

Original

- **Inputs:** $(\log(S/K), \tau/T)$
- **Training:** Fixed $K = 0.9, \sigma = 0.4$
- **Paths:** 100k
- **Pricing:** Learned premium

Interpolation

Inside trained range:
 $K \in [0.7, 1.1], \sigma \in [0.2, 0.6]$

Max RMSE gap across all 9 test scenarios
 ≤ 0.00018

Conditioned

- **Inputs:** $(\log(S/K), \tau/T, K, \sigma)$
- **Training:** $K \sim U(0.7, 1.1), \sigma \sim U(0.2, 0.6)$
- **Paths:** 200k
- **Pricing:** Analytic BS premium

Extrapolation

Outside trained range:
 $K \in \{0.6, 1.3\}, \sigma \in \{0.1, 0.8\}$

Strike RMSE stays within **0.00004**
Worst vol. gap at $\sigma = 0.1$: $\approx 3 \times$ **Optimal**

Source: Submitted report, Sec. 4.1.3, Table 4.3.

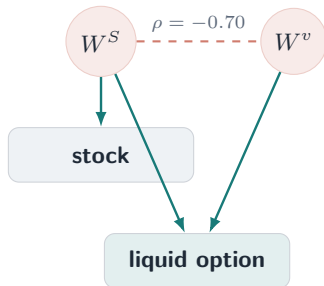
Heston introduces a second source of risk

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{v_t}S_t dW_t^S, \\dv_t &= \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t^v, \\dW_t^S dW_t^v &= \rho dt.\end{aligned}$$

$$dW_t^v = \rho dW_t^S + \sqrt{1 - \rho^2} dW_t^\perp$$

- ▶ v_t is instantaneous variance; $\sqrt{v_t}$ is volatility.
- ▶ Since $|\rho| < 1$, an orthogonal variance shock remains.
- ▶ Stock supplies one traded diffusion exposure; a variance-sensitive liquid option supplies a second locally independent exposure.

Source: Submitted report, Secs. 2.1.2 and 4.1.4; Heston (1993). Feller calculation in backup.



Scope: Local spanning holds in the ideal continuous-time frictionless model; discrete trading still leaves hedging error.

How Heston training paths are generated



$$Z_n^v = \rho Z_n^S + \sqrt{1 - \rho^2} Z_n^\perp, \quad (\delta, \eta) \longrightarrow G_T \longrightarrow HE.$$

Path construction

- ▶ log-Euler updates stock;
- ▶ full-truncation Euler updates variance, so negative Euler values do not enter the square root or next update;
- ▶ the liquid option is repriced at every hedge date.

Training sample

- ▶ train, validation and test paths are simulated independently;
- ▶ one sample is an entire trajectory, not one isolated observation;
- ▶ $S, v, C^h : B \times (N + 1), x : B \times N \times 6$.

Source: Submitted report, Listing C.4 and Algorithms A.3.3, A.6.8, A.6.10.

The hedge chooses two traded positions

$$G_T = \sum_{n=0}^{N-1} \delta_n \Delta S_n + \sum_{n=0}^{N-1} \eta_n \Delta C_n^h$$

δ_n, η_n are desired **total holdings** over $[t_n, t_{n+1}]$, not trades.

Target state	$\log(S_n/K), (T - t_n)/T$
Variance state	v_n/θ
Hedge instrument	$\log(S_n/K_h), (T_h - t_n)/T_h, C_n^h/S_n$

Signed, bounded holdings

$$\delta_n = 5 \tanh(z_{1,n}), \quad \eta_n = 5 \tanh(z_{2,n}).$$

- ▶ Both holdings may be long or short, so $[0, 1]$ is inappropriate.
- ▶ $[-5, 5]$ stabilises training; it is not an optimality theorem.

Two different contracts

target: $K = 0.9, T = 0.5,$

hedge: $K_h = 1, T_h = 1.$

Near ATM and longer dated: variance-sensitive and alive throughout.

Boundary: Full-information simulation: true v_n enters the state; instantaneous variance is latent in real markets.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10.

Frozen-volatility repricing is inconsistent

$$C_t^h \approx C_{\text{BS}}(S_t, K_h, T_h - t, \sqrt{v_t})$$

at t_n : insert
 $\sigma = \sqrt{v_{t_n}}$



price as if σ
stays fixed to T_h



at t_{n+1} : repeat
with new variance

Recomputing the level at every date does not make the path satisfy the Heston pricing equation.

Why it seemed reasonable

- ▶ cheap to evaluate;
- ▶ responsive to current variance;
- ▶ creates a variance-sensitive second instrument.

Process mismatch

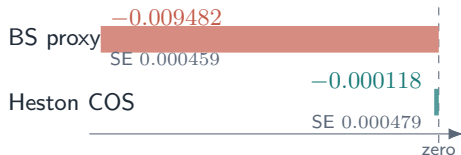
- ▶ stock and variance follow Heston;
- ▶ the proxy uses frozen-volatility Black–Scholes valuation;
- ▶ final COS results use a different traded-option process.

Source: Submitted report, Sec. 4.1.4 (pp. 24–25) and Algorithm A.6.5.

A martingale diagnostic exposes the inconsistency

Estimated mean total liquid-option movement over the target hedge horizon $[0, T]$

$$\hat{m} = \frac{1}{M} \sum_{i=1}^M \left(C_T^{h,(i)} - C_0^h \right), \quad T = 0.5 < T_h = 1.$$



- ▶ At $r = 0$, a consistently risk-neutral-priced traded option has approximately zero mean movement—not a flat path.
- ▶ The proxy estimate is about 20 standard errors from zero.
- ▶ The COS estimate is statistically negligible, within ordinary simulation and discretisation error.

The proxy is not just imprecise in level: its path dynamics are inconsistent with the simulated Heston market.

Source: Submitted report, Table A.9 and Sec. 4.1.4. Diagnostic evidence, not a standalone arbitrage proof.

Why the frozen-volatility proxy drifts

Repriced proxy: $g(t, S, v) = C_{BS}(S, K_h, T_h - t, \sqrt{v})$

Cancelled by the frozen- v BS PDE

$$g_t + rSg_S + \frac{1}{2}vS^2g_{SS} - rg = 0$$

Recomputing the BS level at each date does not make its path satisfy the Heston pricing equation.

1 mean reversion $\times$ variance sensitivity

2 volatility-of-volatility $\times$ variance curvature

Remaining because variance follows Heston

$$R(t, S, v) = \kappa(\theta - v)g_v + \frac{1}{2}\xi^2vg_{vv} + \rho\xi vSg_{Sv}$$

At submitted $r = 0$, discounted and undiscounted drift residuals coincide.

3 spot–variance interaction

Scope: The sign is state-dependent. Under the submitted parameters and simulated states, the average residual was negative, consistent with the diagnostic.

Source: Submitted report, Sec. 4.1.4 and Appendix A.6.3.

COS evaluates a Heston-consistent option price

$$C^H(t, S, v) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T_h - t)}(S_{T_h} - K_h)^+ \mid S_t = S, v_t = v]$$

COS evaluates this conditional expectation efficiently at every simulated state.

Pricing and validation roles

COS	Heston-consistent prices and sensitivities
Carr–Madan	Independent price validation
Finite differences	Sensitivity validation

$$5.1 \times 10^{-8}$$

maximum COS–Carr–Madan price discrepancy

Local analytic position rule

$$\eta_n = \frac{f_v}{g_v}, \quad \delta_n = f_S - \eta_n g_S$$

The option matches variance exposure; the stock removes remaining stock exposure.

Heston COS delta–vega benchmark: Heston COS sensitivities.
BS-proxy Greeks on COS option path: frozen- v BS sensitivities.

Same traded COS option path; different position rule.

Source: Submitted report, Sec. 4.1.4 and Appendices A.6.1–A.6.2. Validation covers the submitted traded range, not global accuracy.

The NN lowers error in all three submitted seeds

Same traded COS option path; different position rule.

Representative strategy	RMSE	Loss CVaR95
No hedge	0.190 061	0.492 959
BS-proxy stock delta	0.019 997	0.046 360
Heston COS delta-vega	0.014 782	0.033 251
Stock-only NN	0.018 166	0.041 379
BS-proxy Greeks on COS option path	0.009 874	0.023 130
Stock + option NN	0.007 426	0.016 624

Against BS-proxy Greeks on COS option path

Seed	NN	Heuristic	Imp. %
111	0.007 806	0.009 838	20.6
222	0.007 715	0.009 871	21.8
333	0.007 346	0.009 864	25.5

22.7%

mean RMSE improvement

27.1%

mean Loss CVaR95 improvement

Protocol note: $\pi_j^* = \mathbb{E}_{\text{test}}[H - G_T^j]$, $HE_j = \pi_j^* + G_T^j - H$. Same-test-path centring removes each strategy's mean error; it compares dispersion and tail risk, not fully out-of-sample pricing.

Interpretation: COS improves model-consistent pricing and local sensitivities—not finite-grid terminal-MSE control.

Source: Submitted report, Tables 4.5–4.6 and A.11. Loss CVaR95 is mean seller loss in the worst 5% of test outcomes.

Three conclusions survive the diagnostics

What the experiments establish

1. **Black–Scholes:** the neural strategy recovered the known finite-grid hedge.
2. **Conditioning:** adding the missing contract information repaired the identified generalisation failure.
3. **Heston:** with a consistently priced liquid option, the stock-and-option NN achieved lower terminal error than the strongest tested local heuristic in every submitted seed.

What they do not establish

- ▶ more than three seeds or fully out-of-sample premium estimation;
- ▶ superiority to a derived finite-grid multi-instrument optimum;
- ▶ performance beyond the frictionless simulated Heston setting and submitted information assumptions.

Observable-information check: The ordering survived a causal EWMA variance proxy; the option quote remained contemporaneously observable and variance-informative.

A constrained and carefully validated neural policy can first recover a known solution and then adapt when the state dynamics, information set, and traded instruments become more complex.

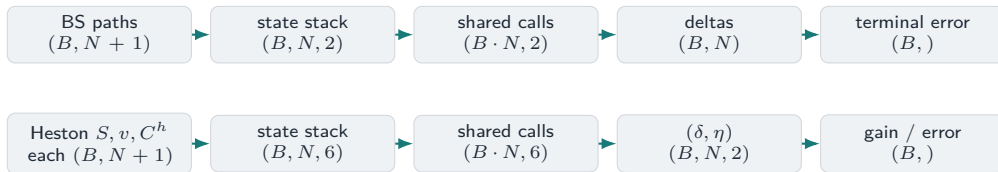
Source: Submitted report, Secs. 4.1.4 and 5; stated limitations and observable-information diagnostic.

Backup: method-to-code map

Mathematical concept	Displayed implementation	Report evidence	Primary expert
Exact BS path simulation	<code>simulate_gbm_paths</code>	Listing C.1; Alg. A.3.1	Micaela / Glasson
Self-financing terminal error	<code>hedge_error</code>	Listing C.2; Alg. A.6.1	Micaela / Glasson
Shared Markov hedge	<code>SharedMarkovHedge</code>	Listing C.3; Sec. 2.2.2	Kaiden / Micaela
Heston full-truncation simulation	PyTorch loop in Listing C.4	Listing C.4; Alg. A.3.3	Glasson / Sipehelele
COS liquid-option pricing	Algorithms A.6.6–A.6.8	App. A.6.1–A.6.2	Glasson
Two-instrument neural gain	Algorithm A.6.10	Sec. 4.1.4	Sipehelele / Glasson

Boundary: The submitted report and its displayed code are present; notebook filenames, result CSV paths and `requirements.txt` are absent from this repository. See `OPEN_QUESTIONS.md`.

State and tensor shapes



The same shared network is applied to every path-date state. Each complete path produces one terminal gain and one terminal hedge error. BS dimensions follow Listings C.1 and C.3; Heston dimensions follow Listing C.4 and Algorithm A.6.10.

Backup: the discrete-time conditional projection

$$\min_{\pi, \delta} \mathbb{E} \left[\left(\pi + \sum_{j=0}^{N-1} \delta_j \Delta S_j - H \right)^2 \right],$$
$$0 = \mathbb{E} \left[\left(H - \pi - \sum_{j=0}^{N-1} \delta_j \Delta S_j \right) \Delta S_n \mid \mathcal{F}_{t_n} \right].$$

Under the submitted $r = 0$ risk-neutral simulation, S is a martingale, so

$$\delta_n^{\text{DT}} = \frac{\mathbb{E}[H \Delta S_n \mid \mathcal{F}_{t_n}]}{\mathbb{E}[(\Delta S_n)^2 \mid \mathcal{F}_{t_n}]} = \frac{\text{Cov}(H, \Delta S_n \mid S_{t_n})}{\text{Var}(\Delta S_n \mid S_{t_n})}.$$

Boundary: This simplification is tied to the submitted $r = 0$ martingale setting; local and global quadratic hedges need not coincide more generally.

Source: Submitted report, Appendix A.4; Schweizer (1995, 2001).

Frozen-volatility drift derivation

For $g(t, S, v) = C_{BS}(S, K_h, T_h - t, \sqrt{v})$, the Heston multiplication rules are

$$(dS)^2 = vS^2 dt, \quad (dv)^2 = \xi^2 v dt, \quad dS dv = \rho \xi v S dt.$$

$$dg = \left[g_t + rSg_S + \kappa(\theta - v)g_v + \frac{1}{2}vS^2g_{SS} + \frac{1}{2}\xi^2vg_{vv} + \rho\xi vSg_{Sv} \right] dt \\ + g_S S \sqrt{v} dW^S + g_v \xi \sqrt{v} dW^v.$$

Frozen- v BS PDE cancellation

$$g_t + rSg_S + \frac{1}{2}vS^2g_{SS} - rg = 0$$

Remaining Heston residual

$$\kappa(\theta - v)g_v + \frac{1}{2}\xi^2vg_{vv} + \rho\xi vSg_{Sv}$$

With submitted $r = 0$, this is both the discounted and undiscounted drift residual. Its sign is state-dependent; the diagnostic is not a complete formal arbitrage proof.

Source: Submitted report, Appendix A.6.3.

COS pricing

$$C_{\text{COS}} \approx e^{-r\tau} \sum_{k=0}^{N_{\text{COS}}-1} 'A_k V_k,$$

$$A_k = \frac{2}{b-a} \Re[\varphi(u_k) e^{-iu_k a}], \quad u_k = \frac{k\pi}{b-a}.$$

A_k approximates the density coefficient; V_k is the analytic call-payoff coefficient. The prime halves the $k = 0$ term.

Finite-difference checks: delta error $\approx 9.8 \times 10^{-5}$; variance-sensitivity error $\approx 4.7 \times 10^{-4}$.

Source: Submitted report, Algorithms A.6.6–A.6.7 and Appendix A.6.2.

- ▶ Heston supplies the characteristic function φ ; COS supplies the numerical expansion.
- ▶ $N_{\text{COS}} = 256$, $[a, b] = [-7, 7]$.
- ▶ Traded validation range:
 $\tau_h \in [0.5, 1]$.
- ▶ Carr–Madan check: $\alpha = 1.5$,
 $u \in [0, 200]$, 20,000 points.

Boundary: Accuracy is validated on the traded grid, not claimed globally.

Analytic Heston stock-and-option hedge

$$df = \cdots + f_S S \sqrt{v} dW^S + f_v \xi \sqrt{v} dW^v,$$
$$d(\delta S + \eta g) = \cdots + (\delta + \eta g_S) S \sqrt{v} dW^S + \eta g_v \xi \sqrt{v} dW^v$$

Match both instantaneous diffusion exposures: $\eta = \frac{f_v}{g_v}, \quad \delta = f_S - \eta g_S.$

Heston COS delta–vega benchmark

Target and hedge-option sensitivities all come from Heston COS.

BS-proxy Greeks on COS option path

Frozen- v BS sensitivities determine positions; gains are still marked to the COS path.

Same traded COS option path; different position rule.

Source: Submitted report, Sec. 4.1.4 and Algorithms A.6.8–A.6.9. Here f_v, g_v are variance sensitivities.

Pricing accuracy and finite-grid hedge optimality are different questions

What COS establishes

- ▶ model-consistent Heston option prices;
- ▶ independently validated prices and implemented sensitivities;
- ▶ local first-order exposures for the COS Greek rule.

What it does not optimise

- ▶ finite-grid terminal MSE;
- ▶ between-date gamma and variance curvature;
- ▶ spot–variance cross effects and changing Greeks.

Approximation-error cancellation may benefit the BS-proxy rule in this tested parameter regime. That ordering does **not** imply that Black–Scholes is a better Heston pricer.

The neural policy instead trains directly on the submitted terminal-MSE objective. Turnover is supporting diagnostic evidence, not proof of the ordering.

Source: Submitted report, Sec. 4.1.4 and the stated interpretation of Tables 4.5–4.6.

Heston simulation

```
for n in range(N_STEPS):
    z1 = torch.randn(n_paths, device=device)
    z2 = torch.randn(n_paths, device=device)
    z_s = z1
    z_v = RHO*z1 + math.sqrt(max(1.0-RHO**2, 0.0))*z2
    v_pos = torch.clamp(v[:, n], min=0.0)
    S[:, n+1] = S[:, n] * torch.exp(
        (R - 0.5*v_pos)*dt + torch.sqrt(v_pos)*sqrt_dt*z_s
    )
    v_next = v[:, n] + KAPPA*(THETA-v_pos)*dt \
        + XI*torch.sqrt(v_pos)*sqrt_dt*z_v
    v[:, n+1] = torch.clamp(v_next, min=0.0)
```

z_s / z_v

stock shock / correlated variance shock

v_{pos}

non-negative variance used in the drift, diffusion and stock log-Euler update

final clamp

prevents a negative Euler proposal becoming the next stored variance

Boundary: The Feller condition $2\kappa\theta = 0.64 \geq \xi^2 = 0.36$ concerns the continuous-time process; it does not guarantee that an Euler proposal is never negative. Train, validation and test paths are simulated independently.

Source: Adapted only for line length from submitted report Listing C.4 and Algorithm A.3.3. Primary speaker: Sipehelele.

Two-instrument gain construction

```
# state at t_n only
x_n = [log(S_n/K), (T-t_n)/T, v_n/THETA,
        log(S_n/K_H), (T_H-t_n)/T_H, C_h_n/S_n]
z_stock, z_option = shared_mlp(x_n)
delta_n = 5.0 * tanh(z_stock)
eta_n = 5.0 * tanh(z_option)

gain = sum(delta_n * (S_next-S_n)
           + eta_n * (C_h_next-C_h_n), over="time")
hedge_error = premium + gain - maximum(S_T-K, 0.0)
loss = mean(hedge_error**2)
```

Boundary: This is a faithful pseudocode rendering of Algorithm A.6.10, not a verbatim excerpt from the absent executable notebook.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10. Primary speakers: Siphelele / Glasson.

Holdings, trades, and gains

Network action: total holdings

$$(\delta_n, \eta_n) \quad \text{over } [t_n, t_{n+1}].$$

Frictionless one-step gain

$$\delta_n \Delta S_n + \eta_n \Delta C_n^h.$$

Trade increments

$$\Delta \delta_n = \delta_n - \delta_{n-1}, \quad \Delta \eta_n = \eta_n - \eta_{n-1}.$$

- ▶ The submitted Heston gain equation uses holdings.
- ▶ Trade increments would determine transaction costs and turnover.
- ▶ Calling δ_n or η_n a trade would therefore change the meaning of the implementation.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10.

Fair-premium centring

$$\pi_j^* = \mathbb{E}_{\text{test}}[H - G_T^j], \quad HE_j = \pi_j^* + G_T^j - H.$$

- removes strategy j 's test-sample mean error;
- RMSE compares centred hedge dispersion;
- Loss CVaR95 compares centred downside tail risk.

Scope: Because π_j^* is estimated on the same test paths as the risk metrics, this is not a fully out-of-sample joint pricing-and-hedging result.

Source: Submitted report, Sec. 4.1.4, Tables A.1–A.2 and Algorithms A.6.1–A.6.2.

Data roles remain separate

Training	Fit weights and learned premium
Validation	Select architecture / best epoch
Test	Final centred risk comparison

Three-seed Heston comparison

Comparator: BS-proxy Greeks on COS option path

Seed	NN RMSE	Heuristic RMSE	RMSE imp. (%)	CVaR95 imp. (%)
111	0.007 806	0.009 838	20.6	26.3
222	0.007 715	0.009 871	21.8	27.0
333	0.007 346	0.009 864	25.5	28.0
Mean pairwise improvement			22.7	27.1

Boundary: Three seeds demonstrate consistency across the submitted runs; they do not support high-powered confidence intervals.

Source: Submitted report, Table A.11.

Why the Heston outputs use scaled tanh

Two signed outputs

$$\delta_n = 5 \tanh(z_{1,n}), \quad \eta_n = 5 \tanh(z_{2,n}).$$

$[-5, 5]$
numerical output range

Why this map is used

- ▶ Stock and option holdings can be long or short.
- ▶ A sigmoid's $(0, 1)$ range is therefore unsuitable.
- ▶ Scaled tanh permits signed, bounded positions.
- ▶ The bound improves numerical stability; it is not an optimality theorem.

Submitted diagnostic: $\max |\eta^{NN}| \approx 1.33$, comfortably below the bound of five.

Source: Submitted report, Sec. 4.1.4, Algorithm A.6.10 and Table A.12.

Observable-information robustness

Causal variance proxy

- ▶ EWMA decay $\lambda = 0.94$;
- ▶ $\text{corr}(\hat{v}_t, v_t) = 0.777613$;
- ▶ relative variance-proxy RMSE = 0.369375.

Submitted mean RMSE

Strategy	RMSE
Stock + option NN	0.007 877
BS-proxy Greeks on COS option path	0.010 734

Scope: C_t^h/S_t remains contemporaneously observable and variance-informative. This is therefore not a stock-return-only filtering experiment.

Source: Submitted report, Sec. 4.1.4 and Tables A.10, A.12, A.15.

Why the hedge option differs from the target

Target liability

$$K = 0.9, \quad T = 0.5$$

Liquid hedge option

$$K_h = 1, \quad T_h = 1$$

- ▶ Near-ATM options generally have useful variance sensitivity; $K_h = 1$ is initially at the money.
- ▶ $T_h > T$ keeps the hedge instrument alive throughout the target horizon.
- ▶ Trading an identical target option would largely trivialise the hedging problem.
- ▶ In practice, the target liability need not itself be a liquid traded contract.

Source: Submitted report, Sec. 4.1.4 and the target/hedge-option configuration.

What the martingale diagnostic measures

Sample mean over $[0, T]$

$$\hat{m} = \frac{1}{M} \sum_{i=1}^M \left(C_T^{h,(i)} - C_0^h \right).$$

Monte Carlo standard error

$$\widehat{\text{SE}}(\hat{m}) = \frac{s(C_T^h - C_0^h)}{\sqrt{M}}.$$

Boundary: Proxy: -0.009482 (SE 0.000459); COS: -0.000118 (SE 0.000479). The COS residual is negligible, not exactly zero.

Source: Submitted report, Table A.9 and Algorithm A.6.8.

- ▶ Individual option paths fluctuate; the statistic concerns their average movement.
- ▶ Under $r = 0$, the risk-neutral reference value is zero.
- ▶ The target horizon ends at $T = 0.5$, before the hedge option matures at $T_h = 1$.
- ▶ The test diagnoses dynamic inconsistency; it is not a standalone proof of executable real-market arbitrage.

Backup: transaction costs change the learned policy

$$V_T^\lambda = \pi + \sum_{n=0}^{N-1} \delta_n \Delta S_n - \lambda \sum_{n=0}^{N-1} S_{t_n} |\delta_n - \delta_{n-1}|, \quad \delta_{-1} = 0.$$

λ	Strategy	RMSE	Loss CVaR95	Turnover
0.0025	BS cost-adjusted	0.008 492	0.021 464	4.497 364
0.0025	Cost-aware NN	0.008 077	0.018 694	4.386 880
0.0050	BS cost-adjusted	0.010 013	0.025 565	4.497 364
0.0050	Cost-aware NN	0.008 331	0.019 269	4.284 602

Boundary: Comparator is a cost-adjusted frictionless delta, not a full optimal-cost benchmark such as Leland or Whalley–Wilmott.

Source: Submitted report, Appendix A.7 and Table A.16.

Backup: data-generation comparison

Paths	Method	RMSE	Loss CVaR95
10,000	Crude MC	0.009 430	0.021 266
10,000	LHS	0.009 201	0.020 845
10,000	Sobol + Brownian bridge	0.009 174	0.020 865
100,000	Crude MC	0.007 865	0.018 218
100,000	LHS	0.007 846	0.018 331
100,000	Sobol + Brownian bridge	0.007 840	0.018 125

At 10,000 paths, structured methods reduce RMSE by roughly 2.4–2.7%; by 100,000 paths the difference is negligible.

Source: Submitted report, Sec. 3.1.5 and Tables A.7–A.8.

Backup: experimental protocol

Experiment	Train	Val	Test	Batch	Training / seeds
Architecture selection	100k	20k	50k	256	60 epochs; 2026–2028
Final BS benchmark	100k	30k	100k	4096	250 epochs; base 123
Parameter robustness	50k	15k	50k	4096	150 epochs
Conditioned hedger	200k	30k	50k / case	4096	250 epochs; 777
Heston stock + option	100k	25k	100k	4096	150 epochs; base 123

- ▶ Adam learning rate 10^{-3} unless otherwise stated.
- ▶ Baseline: $S_0 = 1, K = 0.9, T = 0.5, r = 0, \sigma = 0.4, N = 125$.
- ▶ Architecture search uses more updates per epoch via batch size 256; final experiments use 4096.

Source: Submitted report, Appendix A.1, Table A.1.

Backup: AI assistance and verification

Assistance area disclosed	Human verification reported
Code drafting and debugging	Executed notebooks, code inspection and exported results
Mathematical explanations and LaTeX	Checked against theory and cited sources
Report organisation and consistency	Tables checked against CSV/TeX outputs and figures
COS implementation support	Independent Carr–Madan Fourier price check
Neural experiment support	BS analytic and finite-grid benchmarks; archived outputs
Documentation and reproducibility	Checklists, packaging review and author decisions

Boundary: The report names ChatGPT and Claude. It states that AI suggestions were not authoritative and that all final methodological and interpretive decisions remained with the authors.

Source: Submitted report, Appendix B.

Backup: architecture selection by seed

Seed	Validation winner	Val. loss	Test RMSE
2026	Shared norm. 64×3 , tanh	6.025×10^{-5}	0.007 883
2027	Shared norm. 64×3 , tanh	6.164×10^{-5}	0.007 916
2028	Shared norm. 64×2 , ReLU	6.014×10^{-5}	0.007 855

The 64×3 tanh model won two seeds and had the best mean validation loss and test RMSE; the ReLU alternative was close.

Source: Submitted report, Tables A.3–A.6.

Backup: references I

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